

Bookstore API Test Scenario

Objective:

Implement a RESTful API for a bookstore that allows users to perform CRUD operations on books and implement a complex logical operation for managing book discounts. The API should be built with the Express Framework and Typescript. Please make use of an n-layered architecture.

Tasks:

1. Implement CRUD operations for managing books.
2. Implement a logical operation to calculate the total discounted price for books in a specific genre.

Task Details:

Task 1: CRUD Operations

- Create: Implement an endpoint to add a new book to the inventory.
- Read: Implement an endpoint to retrieve details of a book by its ID.
- Update: Implement an endpoint to update details of an existing book.
- Delete: Implement an endpoint to delete a book from the inventory.

Task 2: Calculate Discounted Price for a Genre

Endpoint: GET /books/discounted-price?genre={genre_name}&discount={discount_percentage}

Description: Calculate the total discounted price for all books in a specific genre based on a given discount percentage.

Example Request:

```
JavaScript
GET /books?genre=Fiction
```

Example Response:

JavaScript

```
[
  {
    "id": 1,
    "title": "To Kill a Mockingbird",
    "author": "Harper Lee",
    "genre": "Fiction",
    ...
  },
  {
    "id": 2,
    "title": "1984",
    "author": "George Orwell",
    "genre": "Fiction",
    ...
  }
]
```

Example Request:

JavaScript

```
GET /books/discounted-price?genre=Fiction&discount=10
```

Example Response:

JavaScript

```
{
  "genre": "Fiction",
  "discount_percentage": 10,
  "total_discounted_price": 112.50
}
```

Calculation Example:

- Suppose there are two books in the "Fiction" genre priced at \$50 and \$75.
- Discount percentage provided is 10%.
- Total original price: $\$50 + \$75 = \$125$.
- Total discounted price: $\$125 - (10\% \text{ of } \$125) = \$112.50$.

Additional Requirements:

- Implement proper input validation and error handling.
- Write unit tests to ensure the correctness of the implemented logical operation.

Submission:

Please place your code in a git repository and share it with james@octoco.ltd. The code must contain a readme which is sufficient to get the API running.